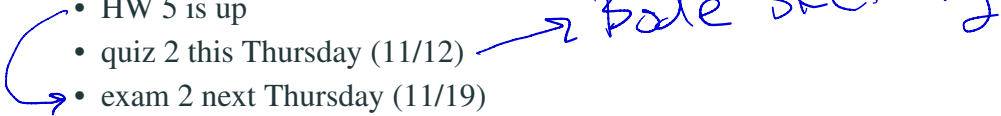
Bode Wrap-Up

EGR 345

GVSU

Announcements

- end of semester schedule posted
- HW 5 is up
- quiz 2 this Thursday (11/12)
- exam 2 next Thursday (11/19)
  - exam 2 topics posted

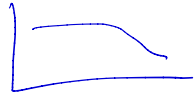


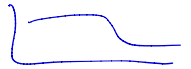
Conceptual Review

- what is a Bode plot?
- where do Bode plots come from?
- what is the key assumption behind Bode plots/frequency response?

dB Mag and phase vs. freq

see next slide





f

↳ TF must be stable → transients die out

→ shows the response to a sinusoidal input over a wide range of input frequencies

Frequency Response

- what do we mean by frequency response?
  - what do we mean by response?

↳ output (t) for a certain input

→ find $y_{ss}(t)$ for sinusoidal input

y_{ss} ↑ after transients die out

ss = steady-state

Frequency Response

- what do we mean by frequency response?
  - what do we mean by response?

$$G(s) = \frac{Y(s)}{U(s)}$$

$M = |G(j\omega)|$

$\phi = \angle G(j\omega)$

Bode

- $u(t) = ?$ $A \sin(\omega t)$
- $y(t) = ?$ $A \cdot M \sin(\omega t + \phi)$

from Bode plot or Laplace s.s. analysis

$G(s) = \frac{Y(s)}{U(s)}$ $\quad u(t) = \sin(\omega t)$

* $G(s)$ is stable

$$U(s) = \frac{\omega}{s^2+\omega^2}$$

$$Y(s) = G(s) \cdot \frac{\omega}{s^2+\omega^2}$$

$$Y(s) = \frac{C_1 \omega}{s^2+\omega^2} + \frac{C_2 s}{s^2+\omega^2} + \text{OTHER TERMS}$$

Transients

$$\left(Y(s) = \frac{C_1 \omega}{s^2+\omega^2} + \frac{C_2 s}{s^2+\omega^2} + \text{OTHER TERMS} \right)(s^2+\omega^2)$$

$$Y(s)(s^2+\omega^2) = C_1 \omega + C_2 s + (\text{OTHER TERMS})(s^2+\omega^2)$$

$$\updownarrow$$

$$Y(s) = G(s) \cdot \frac{\omega}{s^2+\omega^2} \rightarrow Y(s)(s^2+\omega^2) = G(s)\,\omega$$

$$G(s)\,\omega = C_1\,\omega + C_2 s + (\text{OTHER TERMS})(s^2\omega^2)$$

$$(-\omega^2\omega^2) \;\Big]_{s=j\omega}$$

$$G(j\omega)\,\omega = C_1\omega + C_2 j\omega$$

$$C_1 = \text{real}(G(j\omega))$$

$$C_2 = \text{imag}(G(j\omega))$$

$$y_{ss}(t) = C_1 \sin(\omega t) + C_2 \cos(\omega t)$$

$$= M \sin(\omega t + \phi)$$

$$= M \cos(\phi) \sin(\omega t) + M \sin\phi \cos(\omega t)$$

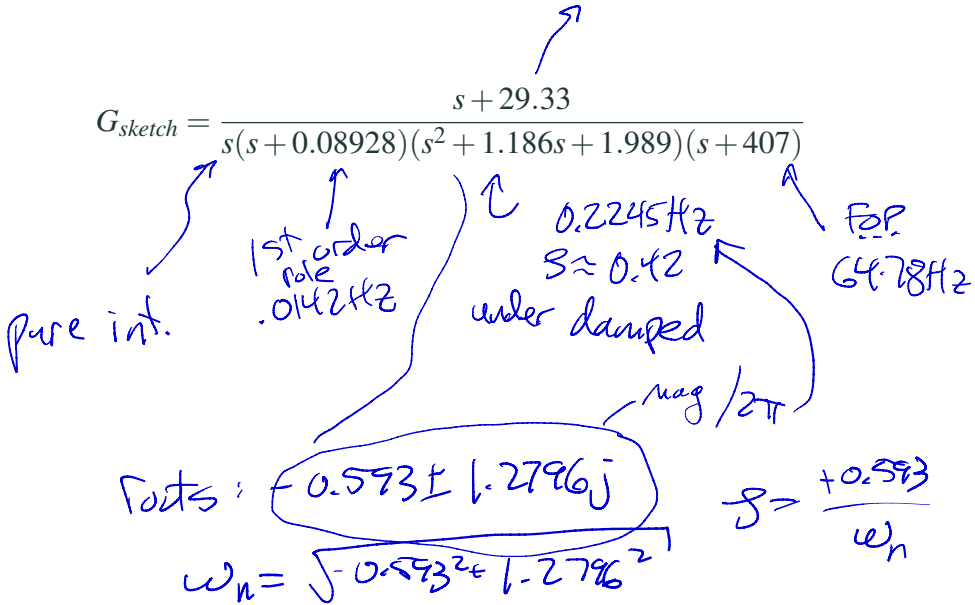
$$\tan\phi = \frac{C_2 \leftarrow \text{imag}}{C_1 \leftarrow \text{real}} \rightarrow \phi = \angle(G(j\omega))$$

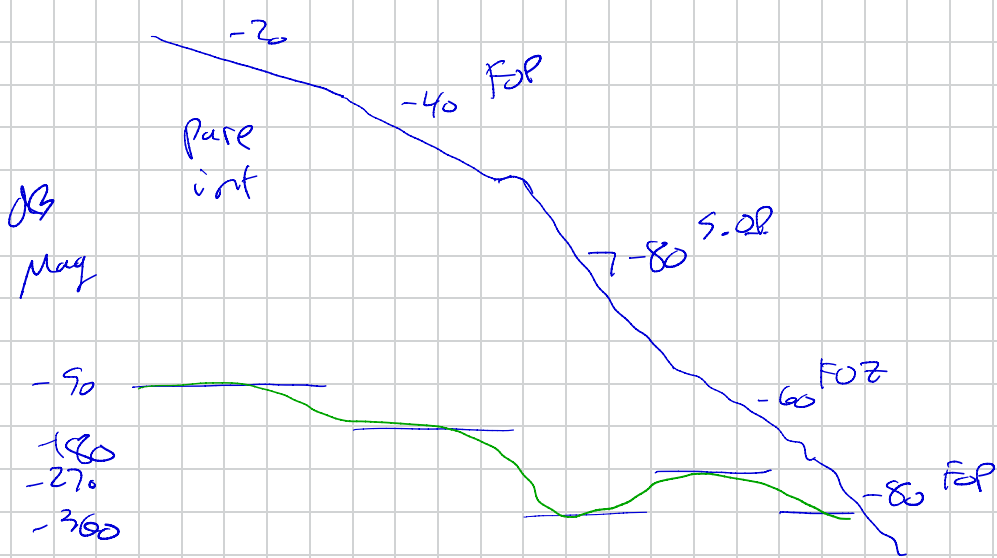
$$M = \sqrt{C_1^2 + C_2^2} = |G(j\omega)|$$

How to make up practice freq. resp. questions and check them:

1. make up a TF
   - or get a random TF from rbg
2. pick an input frequency: $u(t) = \sin(\omega t)$
3. find $y_{ss}(t) = M\sin(\omega t + \phi)$
4. overlay with y_{full} from control.forced_response

Sketching Review



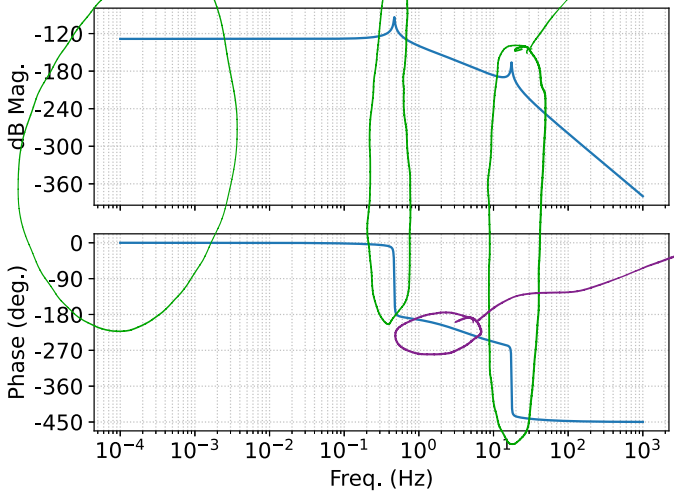

-20
-40
FOP
Pure
int
dB
Mag
-60
-80
S.OP
-90
FOZ
-180
-270
-360
-80
FOP

Sketching Answer Check

```
num = [ 1. ,29.329]
den = [  1. ,408.308,521.248 ,853.055, 72.29, 0.]
G = control.TransferFunction(num,den)
```

System ID Review

- find the TF



System ID Answer Check

```
bode_id_001_11_09_2026_07_49PM.csv
```